

Theory of Computation (Automata Theory and Formal Languages)

Theory of Computation

- mathematical study of computing machines and their capabilities

Strings of symbols – model for representing data that computers manipulate

Alphabets and Languages

An **alphabet** is a finite set of symbols.

Ex. Roman alphabet = $\{a, b, c, \dots, z\}$

Binary alphabet = $\{0, 1\}$

A **string over an alphabet** is a finite sequence of symbols from an alphabet.

Ex. “computer” is a string over the Roman alphabet

01011010 is a string over the binary alphabet

ϵ – denotes the string with no symbols, the empty string

Remarks:

- Strings are usually denoted by u, v, x, y, z and Greek letters
- An alphabet is usually denoted by Σ
- Σ^* is the set of all strings including ϵ over Σ
- the **length** of a string w is its length as a sequence (number of sequence in w), the length of w is denoted by $|w|$

ex. $|\text{computer}| = 8$

$|010110| = 6$

$|\epsilon| = 0$

The **concatenation** of strings x and y , denoted by $x \bullet y = xy$ is the string x followed by the string y .

Ex. $\text{star} \bullet \text{ring} = \text{starring}$

$011 \bullet 100 = 0110$

Some properties:

- $w \bullet \epsilon = w = \epsilon \bullet w$ for any w
- $w \bullet (x \bullet y) = (w \bullet x) \bullet y$

Notes taken from: Elements of the Theory of Computation by Lewis and Papadimitriou

Theory of Computation (Automata Theory and Formal Languages)

A string v is a **substring** of w iff there are strings x and y such that $w = xvy$

If $w = xv$ then x is called a **prefix** of v while v is called a **suffix** of x .

For each string w and each natural number i , the string w^i is defined as

$$w^0 = e$$

$$w^{i+1} = w^i \bullet w \text{ for each } i \geq 0$$

{ their definition is called a definition by induction or recursive definition }

The **reversal** of a string w , denoted by w^R is the string spelled backwards.

$$\text{Ex. } 001^R = 100$$

Formally:

- if $|w| = 0$, then $w^R = w = e$
- if $|w| = n + 1 > 1$ then $w = ua$ for some $a \in \Sigma$ and $w^R = au^R$

Note that for any symbol $a \in \Sigma$, $a^R = a$

Definition:

Any set of strings over an alphabet Σ , i.e. a subset of Σ^* is called a language

- Σ^* , $\emptyset$, Σ are languages
- languages are usually specified by listing all its strings (Since a language is just a set)

Ex.

$$\Sigma = \{0, 1\}$$

$$L_1 = \{0, 01, 011, 0111, \dots\}$$

= set of strings starting with 0 followed by 0 or more 1's

$$L_2 = \{w \in \{0,1\}^* \mid w \text{ has an equal number of 0's and 1's}\}$$

$$= \{01, 1010, 0011\}$$

$$L_3 = \{w \in \{0,1\}^* \mid w = w^R\}$$

= set of palindromes

Theory of Computation (Automata Theory and Formal Languages)

Languages can be specified by

- $L = \{ w \in \Sigma^* \mid w \text{ has property } P \}$
- Since languages are sets, they can be combined by the set operations $\cup$, $\cap$, $-$
- If Σ is understood, we shall write $A' = \Sigma^* - A$
- If L_1 and L_2 are languages over Σ , their **concatenation**, denoted by $L_1 \bullet L_2 = L_1 L_2$ is defined as $L_1 L_2 = \{ w \mid w = xy \text{ for some } x \in L_1 \wedge y \in L_2 \}$

Ex. $\Sigma = \{0,1\}$

$L_1 = \{ w \in \Sigma^* \mid w \text{ has an even number of } 0\text{'s} \}$

$L_2 = \{ w \in \Sigma^* \mid w \text{ starts with a } 0 \text{ followed by } 0 \text{ or more } 1\text{'s} \}$

Then

$L_1 \bullet L_2 = \{ w \in \Sigma^* \mid w \text{ has an odd number of } 0\text{'s in the beginning, followed by } 0 \text{ or more } 1\text{'s} \}$

Note that, if we have an alphabet Σ , then, if $L = \Sigma^*$, then L is the “most general” language that can be formed from the alphabet Σ .

The **Closure (or Kleene Star)** of L denoted by L^* is the set of all strings obtained by concatenating zero or more strings from L .

Thus $L^* = \{ w \in \Sigma^* \mid w = w_1 w_2 \dots w_K, \text{ for some } K \geq 0 \text{ and some } w_1, w_2, \dots, w_K, \text{ ofr some } K \geq 0 \text{ and some } w_1, w_2, \dots, w_K \in L \}$

Ex. 1 $L = \{ 01, 1, 100 \}$ then $110001110011 \in L^*$

Since $1 \bullet 100 \bullet 01 \bullet 1 \bullet 100 \bullet 1 \bullet 1$

Ex. 2 $\emptyset^* = \{ e \}$

→ Concatenation of the zero strings

Note that, for any L over Σ , $e \in L^*$

Theory of Computation (Automata Theory and Formal Languages)

Ex. 3 Show that if $L = \{ w \in \{0,1\}^* \mid w \text{ has unequal number of 0's and 1's} \}$, then $L^* = \{0,1\}^*$

→ Note that if $L_1 \subseteq L_2$, then $L_1^* \subseteq L_2^*$
 $\{0,1\} \subseteq L$, since 0 and 1 regarded as strings has unequal number of 0's and 1's.

Hence $\{0,1\}^* \subseteq L^*$

→ But $L^* \subseteq \{0,1\}^*$ by definition → a language is a subset of Σ^* and in this case $\Sigma = \{0,1\}$

Definition of L^+

$$L^+ = LL^*$$

$$L^+ = \{ w \in \Sigma^* \mid w = w_1w_2\dots w_K \text{ for some } K \geq 1 \text{ and some } w_1w_2\dots w_K \in L \}$$

Question: Is ϵ be an element of L^+

Theorem: For any set L , $L^* = L^{**}$

Proof:

- Every string in L^{**} is made up of strings from L^*
- Every string from L^* is made up of strings from L
- Therefore, every string in L^{**} is made up of strings from L
- Therefore, every string in L^{**} is also a string in L^* that is $L^{**} \subseteq L^*$
- In general, for any set L , it is known that $L \subseteq L^*$, which means therefore that $L^* \subseteq L^{**}$,

Therefore, $L^* = L^{**}$

Theory of Computation (Automata Theory and Formal Languages)

Problem: How to represent language by finite specifications

Regular Expressions (r.e.)

The regular expressions over an alphabet Σ are the strings over the alphabet $\Sigma \cup \{ \}, (\bullet, \emptyset, +, * \}$

1. $\emptyset$ and each member of Σ is a r.e.
2. if α and β are r.e., so is $(\alpha \bullet \beta)$
3. If α and β are r.e. so is $(\alpha + \beta)$
4. If α is a r.e., so is α^* or $(\alpha)^*$
5. Nothing is a regular expression unless it follows from (1) to (4)

• If α is any regular expression then $L(\alpha)$ is the language represented by α . L is a function from strings to languages such that

1. $L(\emptyset) = \emptyset$ and $L(a) = \{a\}$ for all $a \in \Sigma$
2. If α and β are r.e. then
 $L((\alpha \bullet \beta)) = L(\alpha \bullet \beta) = L(\alpha) L(\beta)$
3. if α and β are r.e then
 $L(\alpha + \beta) = L(\alpha) \cup L(\beta)$
4. If α is a r.e. then
 $L(\alpha^*) = L(\alpha)^*$

Ex. 1. $\Sigma = \{a, b\}$, a r.e. over Σ is $((a+b)^*a)$

What is $L(((a+b)^*a))$

$$\begin{aligned} L(((a+b)^*a)) &= L((a+b)^*L(a)) = L((a+b)^*\{a\}) \\ &= L((a+b)^*\{a\}) = (L(a) \cup L(b))^* \{a\} \\ &= (\{a\} \cup \{b\})^* \{a\} \\ &= \{a, b\}^* \{a\} \\ &= \{a, b\}^* a \\ &= \{w \in \{a, b\}^* \mid w \text{ ends in } a\} \end{aligned}$$

Theory of Computation (Automata Theory and Formal Languages)

The class of regular languages (regular sets) over an alphabet Σ is the minimal set of languages containing $\emptyset$ and the singleton sets $\{a\}$, for $a \in \Sigma$ and closed under the operations of union, concatenation and Kleene star. That is, the class $\mathcal{R}$ of regular languages is to have the following properties:

- 1) $\emptyset \in \mathcal{R}$ and if $a \in \Sigma$ then $\{a\} \in \mathcal{R}$
- 2) if $A, B \in \mathcal{R}$ then $A+B$, $A \bullet B$ and A^* are members of $\mathcal{R}$
- 3) If S is the set of languages satisfying 1 and 2, then $\mathcal{R} \subseteq S$.

Note:

- A language is regular iff it can be described by a regular expression
- A regular expression may be viewed as a language generator

Examples:

1. $L(a^*b^*) =$ set of all strings in $\{a,b\}^*$ where the a 's come before the b 's.
Is $a^*b^* = (ab)^*$?
 2. $L(a(aa)^*) =$ set of all strings in $\{a\}^*$, the number of a 's is odd.
 3. $(a+b)^*a =$ all strings ending in a .
- Regular expressions are language generators